

Problem Statement

Perform 2 iteration of the gradient descent to get updated weights w_0, w_1, w_2, w_3 .
[Initial $w_0 = w_1 = w_2 = w_3 = 1$ and $\alpha = 0.05$]

x1	x2	x3	y
2	7	6	6.5
4	8	7.5	7.8
6	9	8.5	8.8

I



Linear Regression

Initial weights

$$w_0 = w_1 = w_2 = w_3 = 1$$

learning rate $\alpha = 0.05$

1st iteration
prediction

$$\hat{y} = w_0 + w_1 x_1 + w_2 x_2 + w_3 x_3$$

$$\hat{y}_1 = 1 + 1 \times 2 + 1 \times 7 + 1 \times 6$$

$$= 16$$

$$\hat{y}_2 = w_0 + w_1 x_1 + w_2 x_2 + w_3 x_3$$

$$= 1 + 4 + 8 + 7.5$$

$$= 20.5$$

$$\hat{y}_3 = 1 + 6 + 9 + 8.5$$

$$= 24.5$$

slope / Gradient

$$S_j = \sum (\hat{y} - y) x_j^0$$

calculating slope

For w_0

$x_0 = 1$ for all

$$S_0 = (16 - 6.5) \times 1 + (20.5 - 7.8) \times 1 + (24.5 - 8.8) \times 1$$
$$= 37.9$$

For w_1

$$\text{slope, } S_1 = (16 - 6.5) \times 2 + (20.5 - 7.8) \times 4 + (24.5 - 8.8) \times 6$$
$$= 164$$

For w_2

$$\text{slope, } S_2 = (\hat{y}_1 - y_1) x_1 + (\hat{y}_2 - y_2) x_2 + (\hat{y}_3 - y_3) x_3$$
$$= (16 - 6.5) \times 7 + (20.5 - 7.8) \times 8 + (24.5 - 8.8) \times 9$$
$$= 309.4$$

For w_3

$$\begin{aligned}\text{slope, } s_3 &= (\hat{y} - y)x_3 + (\hat{y} - y)x_3 + (\hat{y} - y)x_3 \\ &= (16.6 - 5)6 + (20.5 - 7.8)7.5 + (24.5 - 8.8)8.5 \\ &= 285.7\end{aligned}$$

update weight

$$w'_0 = w_0 - a s_0$$

$$= 1 - 0.05 \times 37.9$$

$$= -0.895$$

$$\begin{aligned}w'_1 &= w_1 - a s_1 = 1 - 0.05 \times 164 \\ &= -7.2\end{aligned}$$

$$\begin{aligned}w'_2 &= w_2 - a s_2 = 1 - 0.05 \times 309.4 \\ &= -14.47\end{aligned}$$

$$\begin{aligned}w'_3 &= w_3 - a s_3 = 1 - 0.05 \times 285.7 \\ &= -13.285\end{aligned}$$

After 1st Iteration

$$w'_0 = -0.895$$

$$w'_1 = -7.2$$

$$w'_2 = -14.47$$

$$w'_3 = -13.285$$

prediction

$$\hat{y}_1 = w'_0 + w'_1 x_1 + w'_2 x_2 + w'_3 x_3$$

$$= -0.895 - 7.2 \times 2 - 14.47 \times 7 - 13.285 \times 6$$

$$= -196.295$$

$$\hat{y}_2 = ~~w'_0~~ + w'_0 = -0.895 - 7.2 \times 4 - 14.47 \times 8 - 13.285 \times 7.5$$
$$= -249.078$$

$$\hat{y}_3 = -0.895 - 7.2 \times 6 - 14.47 \times 9 - 13.285 \times 8.5$$
$$= -287.2475$$

Slope for a weight w_0
 $x = 1$ for all

$$S_0' = \sum (\hat{y} - y) x_0 \\ = (-196.295 - 6.5) \times 1 + (-249.078 - 7.8) \times 1 + \\ (-287.2475 - 8.8) \times 1$$

$$= -755.72$$

$$S_1' = \sum (\hat{y} - y) x_1 \\ = (-196.295 - 6.5) \times 2 + (-249.078 - 7.8) \times 4 + \\ (-287.2475 - 8.8) \times 6$$

$$= -3209.387$$

$$S_2' = (-196.295 - 6.5) \times 7 + (-249.078 - 7.8) \times 8 + \\ (-287.2475 - 8.8) \times 9$$

$$= -6139.0165$$

$$S_3' = (-196.265 - 6.5) \times 6 + (249.078 - 7.8) \times 7.5 \\ + (-287.2475 - 8.8) \times 8.5$$

$$= -5761.15625$$

updated weight

$$w_0'' = w_0' - aS_0'$$

$$= -0.895 - 0.05 \times (-755.72)$$

$$= 36.891$$

$$w_1'' = w_1' - aS_1'$$

$$= -7.2 - 0.05 \times (-3209.387)$$

$$= 153.269$$

$$w_2'' = w_2' - aS_2' = -14.47 - 0.05 \times (-6139.016)$$

$$= 292.48$$

$$w_3'' = w_3' - a s_3'$$

$$= -13.285 - 0.05 \times (-5761.15625)$$

$$= 274.77$$

Logistic Regression

prediction

$$z_0 = w_0 + w_1 x_1 + w_2 x_2 + w_3 x_3$$

$$= 1 + 2 - 7 + 6$$

$$= 2$$

Activation Function

$$\hat{y}_1 = \frac{1}{1 + e^{-z_0}} = \frac{1}{1 + e^{-2}} = 0.88$$

$$z_1 = w_0 + w_1 x_1 + w_2 x_2 + w_3 x_3$$

$$= 1 + 4 - 8 + 7.5 = 4.5$$

$$\hat{y}_2 = \frac{1}{1 + e^{-z_1}} = \frac{1}{1 + e^{-4.5}} = 0.98$$

$$z_2 = w_0 + w_1 x_1 + w_2 x_2 + w_3 x_3$$

$$= 1 + 6 - 9 + 8.5 = 6.5$$

$$\hat{\gamma}_3 = \frac{1}{1 + e^{-2.3}} = \frac{1}{1 + e^{6.5}} = 0.998$$

slope for w_0

$$S_0 = \sum (\hat{y} - y) x_0$$

for all $x = 1$

$$= (0.881 - 0) \cdot 1 + (0.989 - 1) \cdot 1 + (0.998 - 1)$$

$$= 0.868$$

$$S_1 = (0.881 - 0) \cdot 2 + (0.989 - 1) \cdot 4 + (0.998 - 1) \cdot 6$$

$$= 1.706$$

$$S_2 = (0.881 - 0) \cdot (-7) + (0.989 - 1) \cdot (-8) + (0.998 - 1) \cdot (-9)$$

$$= -6.061$$

$$s_3 = 5.1865$$

updated weight

$$\begin{aligned} w_0' &= w_0 - a s_0 \\ &= 1 - 0.05 \times 0.868 = 0.9566 \end{aligned}$$

$$\begin{aligned} w_1' &= w_1 - a s_1 \\ &= 1 - 0.05 \times 1.706 = 0.9147 \end{aligned}$$

$$\begin{aligned} w_2' &= 1 - a s_2 \\ &= 1 - 0.05 \times (-6.061) \\ &= 1.30305 \end{aligned}$$

$$\begin{aligned} w_3' &= w_3 - a s_3 \\ &= 1 - 0.05 \times (5.1865) \\ &= 0.740675 \end{aligned}$$

$(2.0, 2.0), (2.0, 8.0), (8.0, 0.0), (8.0, 0.0), (0.0, 0.0) - b$
 $(1, 1), (1, 8.0), (8.0, 8.0), (8.0, 2.0), (8.0, 0.0), (8.0, 8.0)$

ROC curve

Total positive (p) = 5

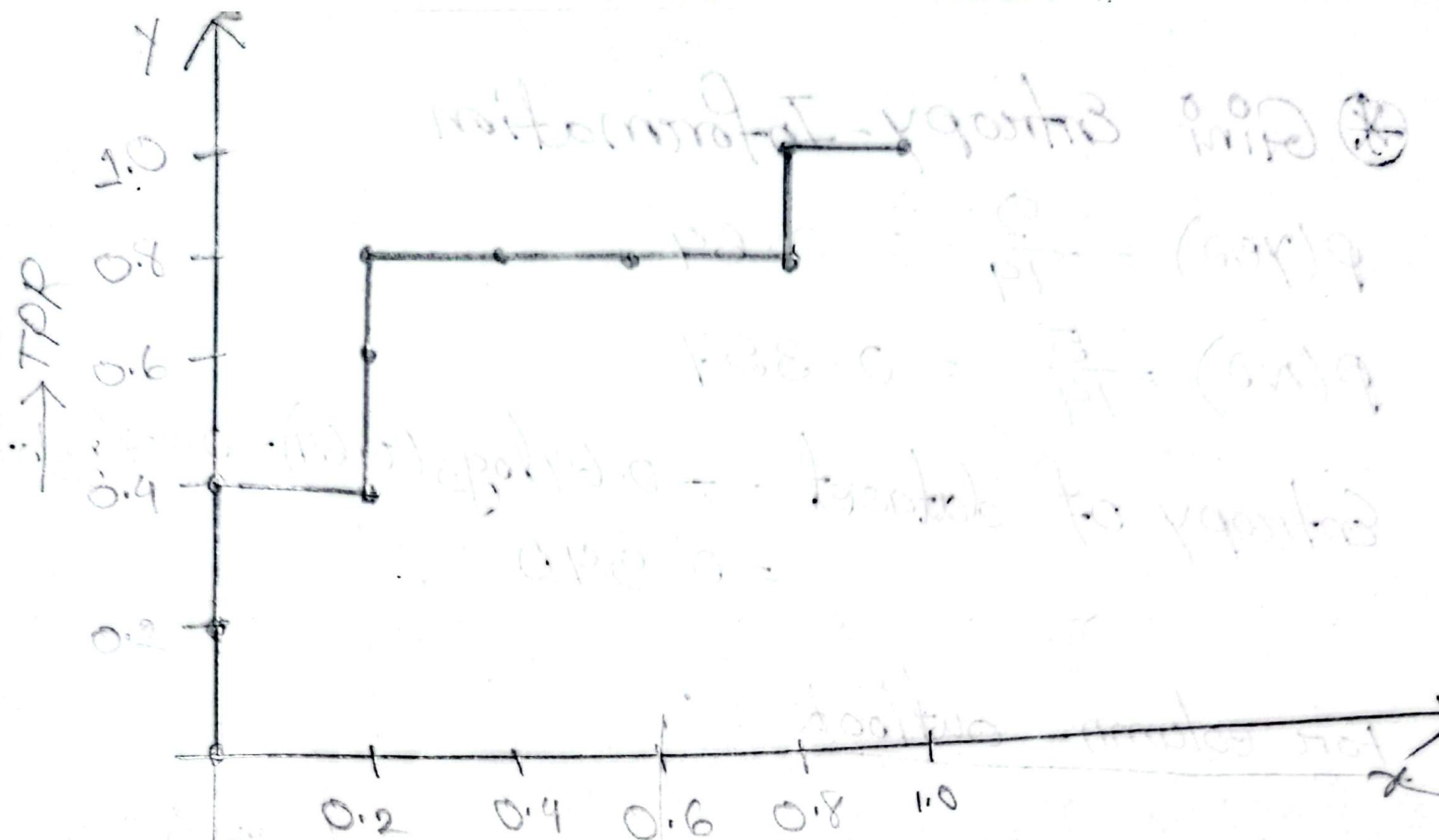
Total negative (n) = 5

$$TPR = \frac{TP}{TP + FN} = \frac{TP}{\text{Total } p}$$

$$FPR = \frac{FP}{FP + TN} = \frac{FP}{\text{Total } n}$$

step	class	TP	FP	TPR = $\frac{TP}{\text{total } p}$	FPR = $\frac{FP}{\text{total } n}$
start	-	0	0	0	0
1	p	1	0	0.2	0
2	p	2	0	0.4	0
3	N	2	1	0.4	0.2
4	p	3	1	0.6	0.2
5	p	4	1	0.8	0.2
6	N	4	2	0.8	0.4
7	N	4	3	0.8	0.6
8	N	4	4	0.8	0.8
9	p	5	4	1.0	0.8
10	N	5	5	1.0	1.0

point - (0, 0), (0, 0.2), (0, 0.4), (0.2, 0.4), (0.2, 0.6), (0.2, 0.8), (0.4, 0.8), (0.6, 0.8), (0.8, 0.8), (0.8, 1), (1, 1)



	FPR	TPR	Classification
True	0	0	True
False	0	1	False
True	1	0	True
False	1	1	False

$$\overline{i-1}^{1/q}$$

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Day	Outlook	Temp.	Humidity	Wind	Decision
1	Sunny	Hot	High	Weak	No
2	Sunny	Hot	High	Strong	No
3	Overcast	Hot	High	Weak	Yes
4	Rain	Mild	High	Weak	Yes
5	Rain	Cool	Normal	Weak	Yes
6	Rain	Cool	Normal	Strong	No
7	Overcast	Cool	Normal	Strong	Yes
8	Sunny	Mild	High	Weak	No
9	Sunny	Cool	Normal	Weak	Yes
10	Rain	Mild	Normal	Weak	Yes
11	Sunny	Mild	Normal	Strong	Yes
12	Overcast	Mild	High	Strong	Yes
13	Overcast	Hot	Normal	Weak	Yes
14	Rain	Mild	High	Strong	No

<https://sefiks.com/2018/08/27/a-step-by-step-cart-decision-tree-example/>

⊗ Gini Entropy-Information

$$P(\text{yes}) = \frac{9}{14} = 0.64$$

$$P(\text{no}) = \frac{5}{14} = 0.357$$

$$\begin{aligned} \text{Entropy of dataset} &= -0.64 \log_2(0.64) - 0.357 \log_2(0.357) \\ &= 0.940 \end{aligned}$$

For column - outlook

outlook	yes	no	instance
sunny	2	3	5
Overcast	4	0	4
Rain	3	2	5

$$\begin{aligned} \text{Sunny} &= -\frac{2}{5} \log_2\left(\frac{2}{5}\right) - \\ &\quad \log_2\left(\frac{3}{5}\right) \\ &= 0.971 \end{aligned}$$

$$\begin{aligned} \text{overcast} &= -\frac{4}{4} \log_2\left(\frac{4}{4}\right) - 0 \\ &= 0 \end{aligned}$$

$$\begin{aligned} \text{Entropy - Rain} &= -\frac{3}{5} \log_2\left(\frac{3}{5}\right) \\ &\quad - \frac{2}{5} \log_2\left(\frac{2}{5}\right) \\ &= 0.971 \end{aligned}$$

weighted Entropy

$$= \frac{5}{14} \times 0.971 + \frac{4}{14} \times 0 + \frac{5}{14} \times 0.971$$

$$= 0.694$$

Information Gain = $0.940 - 0.694$
 $= 0.246$

For column - Temperature

temp	yes	no	inst
Hot	2	2	9
Mild	4	2	6
cool	3	1	4

$$\text{Entropy-Hot} = -\frac{2}{4} \log_2\left(\frac{2}{4}\right) - \frac{2}{4} \log_2\left(\frac{2}{4}\right)$$

$$= 1$$

$$\text{Entropy-mild} = -\frac{4}{6} \log_2\left(\frac{4}{6}\right) - \frac{2}{6} \log_2\left(\frac{2}{6}\right)$$

$$= 0.918$$

$$\text{Entropy-cool} = -\frac{3}{4} \log_2\left(\frac{3}{4}\right) - \frac{1}{4} \log_2\left(\frac{1}{4}\right)$$

$$= 0.811$$

weighted Entropy

$$= \frac{4}{14} \times 1 + \frac{6}{14} \times 0.918 + \frac{4}{14} \times 0.811$$

$$= 0.910$$

Information Gain

$$IG(\text{temp}) = 0.940 - 0.910$$

$$= \boxed{0.029}$$

For Humidity column

Hum	yes	no	inst
High	3	4	7
Normal	6	1	7

$$\text{Entropy-High} = -\frac{3}{7} \log_2 \left(\frac{3}{7} \right) - \frac{4}{7} \log_2 \left(\frac{4}{7} \right)$$

$$= 0.985$$

$$\text{Entropy-Normal} = -\frac{6}{7} \log_2 \left(\frac{6}{7} \right) -$$

$$\frac{1}{7} \log_2 \left(\frac{1}{7} \right)$$

$$= 0.591$$

weighted Entropy of Humidity

$$= \frac{7}{14} \times 0.985 + \frac{7}{14} \times 0.591 = 0.788$$

Gain Information,

$$GI = 0.940 - 0.788$$

$$= \boxed{0.1516}$$

For Wind column

wind	yes	no	inst
week	6	2	8
strong	3	3	6

$$\begin{aligned}\text{Entropy-week} &= -\frac{6}{8} \log_2\left(\frac{6}{8}\right) \\ &\quad - \frac{2}{8} \log_2\left(\frac{2}{8}\right) \\ &= 0.811\end{aligned}$$

$$\begin{aligned}\text{Entropy-strong} &= -\frac{3}{6} \log_2\left(\frac{3}{6}\right) \\ &\quad - \frac{3}{6} \log_2\left(\frac{3}{6}\right) \\ &= 1\end{aligned}$$

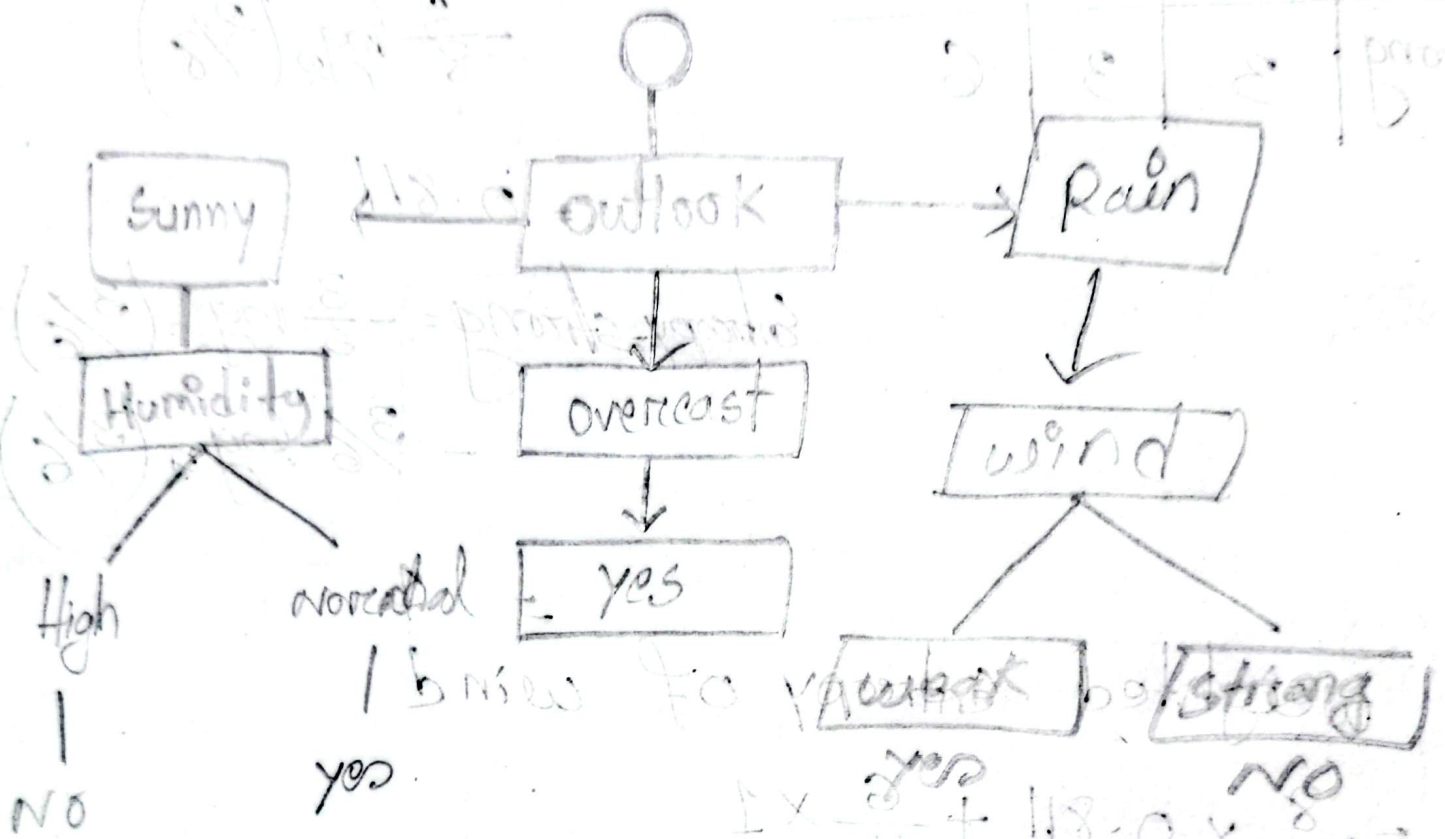
weighted entropy of wind

$$\begin{aligned}&= \frac{8}{14} \times 0.811 + \frac{6}{14} \times 1 \\ &= 0.892\end{aligned}$$

Information Gain

$$IG = 0.94 - 0.892 = \boxed{0.048}$$

Highes - outlook



$$1 \times \frac{2}{11} + 118.0 \times \frac{8}{11} = 90.9$$

$$0.040 = 90.9 - 90.9 = 0$$

look - Sunny & temperature

temperature	yes	no	inst
hot	0	2	2
mild	0	1	2
cool	1	0	1

$$\text{Entropy - hot} = -\frac{2}{2} \log_2(2/2) = 0$$

$$\text{Entropy - mild} = -\frac{1}{2} \log_2(1/2) - \frac{1}{2} \log_2(1/2) = 1$$

$$\text{Entropy - cool} = -\frac{1}{1} \log_2(1/1) = 0$$

weighted temperature

$$= \frac{2}{5} \times 0 + \frac{2}{5} \times 1 + \frac{1}{5} \times 0 = 0.4$$

$$IG_1 = 0.971 - 0.4$$

$$= 0.571$$

look - sunny & Humidity

Hum	yes	no	inst
High	0	3	3
Normal	2	0	2

$$\text{Entropy - High} = -\frac{3}{3} \log_2(3/3) = 0$$

$$\text{Entropy - Normal} = -\frac{2}{2} \log_2(2/2) = 0$$

weighted Humidity

$$= 0$$

$$IG = 0.971 - 0$$

$$= 0.971$$

outlook - sunny & wind

wind	yes	no	inst
weak	1	2	3
strong	1	1	2

$$\text{Entropy-weak} = -\frac{1}{3} \log_2\left(\frac{1}{3}\right) - \frac{2}{3} \log_2\left(\frac{2}{3}\right)$$

$$= 0.918$$

$$\begin{aligned} \text{Entropy-strong} &= -\frac{1}{2} \log_2\left(\frac{1}{2}\right) \\ &\quad - \frac{1}{2} \log_2\left(\frac{1}{2}\right) \\ &= 1 \end{aligned}$$

weighted wind

$$= \frac{3}{5} \times 0.918 + \frac{2}{5} \times 1$$

$$= 0.9508$$

outlook - Rain & temperature

temp	yes	no	inst
hot	0	0	0
mild	2	1	3
cool	1	1	2

$$\text{Entropy - hot} = 0$$

$$\text{Entropy - mild} = -\frac{2}{3} \log_2(2/3) - \frac{1}{3} \log_2(1/3)$$

$$= 0.918$$

$$\text{Entropy} = -\frac{1}{2} \log_2(1/2) - \frac{1}{2} \log_2(1/2)$$

$$= 1$$

weighted temperature

$$= \frac{3}{5} \times 0.918 + \frac{2}{5} \times 1 = 0.9508$$

$$\text{Information Gain} = 0.971 - 0.9508$$

$$= 0.0202$$

outlook - Rain & Humidity

Hum	yes	no	inst
High	1	1	2
Normal	2	1	3

$$\text{Entropy - High} = -\frac{1}{2} \log_2(1/2) - \frac{1}{2} \log_2(1/2)$$

$$= 1$$

$$\text{Entropy - normal} = -\frac{2}{3} \log_2(2/3) - \frac{1}{3} \log_2(1/3)$$

$$= 0.918$$

Weighted Humidity

$$= \frac{2}{5} \times 1 + \frac{3}{5} \times 0.918$$

$$= 0.9508$$

$$IG = 0.971 - 0.9508$$

$$= 0.0202$$

outlook - Rain & Wind

wind	yes	no	inst
weak	3	0	3
strong	0	2	2

Entropy-weak = 0

Entropy-strong = 0

weight wind = 0

$$IG = 0.971 - 0$$

$$= 0.971$$

⑧ confusion matrix (chintan) kosh-8

	TP 80	FN 20
	FP 30	TN 70

1. Accuracy

$$\begin{aligned}\text{Accuracy} &= \frac{TP + TN}{TP + FP + TN + FN} \\ &= \frac{80 + 70}{80 + 30 + 70 + 20} \\ &= 0.75\end{aligned}$$

2. precision

$$\begin{aligned}\text{precision} &= \frac{TP}{TP + FP} \\ &= \frac{80}{80 + 30} \\ &= 0.727\end{aligned}$$

3. Recall (Sensitivity)

$$\text{Recall} = \frac{TP}{TP + FN}$$
$$= \frac{80}{80 + 20} = 0.80$$

4. F1-Score

$$F1 = \frac{2 \times \text{Recall} \times \text{precision}}{\text{precision} + \text{Recall}}$$

$$= \frac{2 \times 0.80 \times 0.727}{0.727 + 0.80}$$
$$= 0.761$$

b. Suppose you have three different classifiers (Decision Tree, SVM, and Logistic Regression) for a classification task. How can you combine their predictions using **bagging** and **boosting**? Explain with proper diagram.

c. A hospital develops a machine learning model to detect cancer from patient scans. Out of 1000 patients: 3

- 100 patients actually have cancer.
- The model predicts 120 patients as having cancer. Among them, 80 actually have cancer.

- i. Calculate the precision and recall of the model.
- ii. Which metric (precision or recall) would you prioritize in a cancer screening system and why?

Q.4 a. Explain the concept of a labeled dataset and its significance in the context of supervised learning algorithms. 2

b. Suppose you are building a Transformer-based model for **machine translation** from English to French: 1+2+1

- Which part of the Transformer processes the English input, and which part generates the French output?
- Why is positional encoding necessary in both encoder and decoder?
- What is the role of Masked Self-Attention in Decoder?

3(c)

Total patients = 1000

Actual cancer patients = 100 $\rightarrow (TP + FN = 100)$

predicted cancer patients = 120 $\rightarrow (TP + FP = 120)$

Among them actually have cancer $TP = 80$

$$FP = 120 - 80 = 40$$

$$FN = 100 - 80 = 20$$

$$TN = 1000 - (TP + FP + FN) = 1000 - 80 - 40 - 20 = 860$$

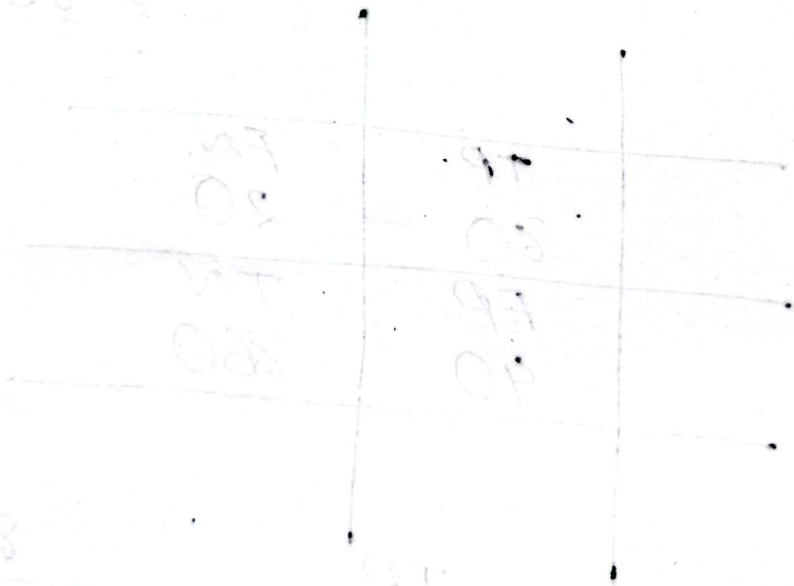
	TP	FN
	80	20
	FP	TN
	40	860

$$\begin{aligned} \text{precision} &= \frac{TP}{TP + FP} = \frac{80}{80 + 40} \\ &= \frac{80}{120} \\ &= 0.67 \end{aligned}$$

$$\text{Recall} = \frac{TP}{TP + FN} = \frac{80}{80 + 20} = 0.8$$

Recall metric is important

In cancer scan screening FN is very dangerous. In case a patient is effected by cancer but predicted as not then cancer patient life is in risk.



⊛ Loss Function - cross entropy

$$\epsilon = \sum (-y \log(\hat{y}) - \{(1-y) \log(1-\hat{y})\}) = m$$

⊛ Linear Regression (Direct formula)

Given dataset

x	y
1	2
2	3
3	5

Find the regression line using direct formula

Solution

mean of x

$$\bar{x} = \frac{1+2+3}{3} = 2$$

mean of y

$$\bar{y} = \frac{2+3+5}{3} = 3.33$$

calculating slope,

$$m = \frac{\sum (y - \bar{y})(x - \bar{x})}{\sum (x - \bar{x})^2}$$

x	y	$\bar{x}$	$\bar{y}$	$x - \bar{x}$	$y - \bar{y}$	multiply	$(x - \bar{x})^2$
1	2	2	3.33	-1	-1.33	1.33	1
2	3	2	3.33	0	-0.33	0	0
3	5	2	3.33	1	1.67	1.67	1
						net: 3	net: 2

$$m = \frac{3}{2} = 1.5$$

Intercept (c)

$$c = \bar{y} - m\bar{x}$$

$$= 3.33 - 1.5 \times 2$$

$$= 0.33$$

$\therefore$ Regression Equation

$$y = mx + c$$

$$= 1.5x + 0.33$$

* confusion matrix

mcc (matthews correlation coefficient)

$$mcc = \frac{TP \cdot TN - FP \cdot FN}{\sqrt{(TP+FP)(TP+FN)(TN+FP)(TN+FN)}}$$

* Adaboost

Alpha

$$\alpha^t = \frac{1}{2} \ln \left(\frac{1 - e^t}{e^t} \right) \quad e^t = \text{error rate}$$

weight update

$$w_i^{t+1} = \frac{w_i^t}{Z} e^{-\alpha^t h^t(x_i) y_i}$$

$h(x) = \text{prediction } (+1/-1)$

$y = \text{actual } (+1/-1)$

Q.6 a. In what scenarios would you recommend using deep learning over machine learning, and vice versa, considering factors like data complexity, computational resources, and interpretability?

b. What is the main difference between **PCA** and **LDA** in terms of their purpose in data analysis? Suppose you have a dataset with 3 classes and 5 features. Explain how **PCA** reduces the dimensionality of the dataset. 1-

c. What is the significance of Confusion Matrix? Consider a confusion matrix: 1+3

Classes	Yes	No
Yes	90	210
No	140	9560

Calculate the following metrics from above confusion matrix:

- i) Specificity
- ii) Sensitivity
- iii) Overall accuracy

6(c) 2025

TP	FN
90	210
FP	TN
140	9560

$$i) \text{ specificity} = \frac{TN}{TN + FP}$$

$$= \frac{9560}{9560 + 140}$$

$$= 0.98$$

$$ii) \text{ sensitivity / Recall} = \frac{TP}{TP + FN}$$

$$= \frac{90}{90 + 210}$$

$$= 0.3$$

$$iii) \text{ Overall Accuracy} = \frac{TP + TN}{\text{Total}}$$

$$= \frac{90 + 9560}{10000}$$

$$= 0.965$$

Premier University
Department of Computer Science & Engineering
Spring 2023, Midterm
CSE 457: Machine learning

$A + B + C$

Total Marks: 20

Time: 40 minutes

Answer any two from the following questions.

Q.1 a) Describe the gradient descent algorithm for linear regression. CO1 05

b) Given below are five observations collected in a simple regression. Study on two variables, x (independent variables) and y (dependent variable). CO1 05

x	y
2	7
4	5
6	4
8	2
10	1

i) Using normal equation, calculate intercept b and slope w and write down the estimated regression equation.

ii) for $x=8$, predict the value of dependent variable, and then calculate the residual.

Q.2 a) Differentiate between gradient descent and normal equation CO1 05

b) Suppose $m=4$ students have taken some class, and the class had a midterm exam and a final exam. You have collected a dataset of their scores on the two exams, which is as follows: CO1 05

Midterm	(midterm	Final
		Exam



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1(b)

(1)

x	y
2	7
4	5
6	4
8	2
10	1

Normal Equation of Linear Regression

$$y = bx + b$$

mean value:

$$\bar{x} = \frac{2+4+6+8+10}{5} = 6$$

$$\bar{y} = \frac{7+5+4+2+1}{5} = 3.8$$

x	y	$\bar{x}$	$\bar{y}$	$x - \bar{x}$	$y - \bar{y}$	multiply	$(x - \bar{x})^2$
2	7	6	3.8	-4	3.2	-12.8	16
4	5	6	3.8	-2	1.2	-2.4	4
6	4	6	3.8	0	0.2	0	0
8	2	6	3.8	2	-1.8	-3.6	4
10	1	6	3.8	4	-2.8	-11.2	16
						$\Sigma -30$	$\Sigma 40$

$$\text{slope, } w = \frac{\sum (x - \bar{x})(y - \bar{y})}{\sum (x - \bar{x})^2}$$

$$= \frac{-30}{40} = -0.75$$

$$\text{intercept, } b = \bar{y} - w\bar{x}$$

$$= 3.8 + 0.75 \times 6$$

$$= 8.3$$

Final Regression Equation,

$$y = wx + b$$

$$= -0.75x + 8.3$$

(ii) predict for $x = 8$

$$y = -0.75(8) + 8.3$$

$$= 2.3$$

$(x - \bar{x})$	$y - \bar{y}$	$(x - \bar{x})^2$	$(x - \bar{x})(y - \bar{y})$
2	2.3	4	4.6
1	2.1	1	2.1
0	2.0	0	0.0
1	1.1	1	1.1
2	0.2	4	0.4
Σ	Σ	Σ	Σ
6	8.3	10	-30